

UI/UX - ADAPTIVE DASHBOARD DESIGN SYSTEM REAL TIME RISK MONITORING

Category: UI/UX design | Adaptive Dashboard Design System

Timeline: 15 Days | AI-Accelerated Innovation & Research Project

Target: Undergraduate / MBA / Engineering Students

TASK:

Design and deliver a production-grade, adaptive dashboard design system and high-fidelity prototype suite for RiskLens Technologies - a fictional fintech startup that has secured USD 25 million in Series B funding and requires a world-class UI/UX system to compete with Bloomberg PORT, MSCI RiskMetrics, and Axioma. The current frontend is a crude wireframe with no design system, zero accessibility compliance, and poor responsiveness across devices. The design system must include a minimum of 40 reusable components with all required states and variants, implement role-based view customisation for four distinct user personas - Junior Risk Analyst, Senior Portfolio Manager, Chief Risk Officer, and Compliance Officer - deliver high-fidelity interactive prototypes across three device breakpoints (desktop 1440px, tablet 768px, mobile 375px) covering a minimum of 12 screens, enforce WCAG 2.1 AA accessibility compliance across all screens, incorporate specialised financial data visualisation components for real-time risk metrics including VaR, CVaR, Greeks, drawdown, and stress test outputs, and produce a complete design handoff specification document enabling the frontend engineering team to implement without ambiguity

PART A: TRAINING MATERIAL & LEARNING RESOURCES

PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

PART C: CASE STUDIES & REAL-WORLD APPLICATIONS

PART D: 15-DAY PROJECT METHODOLOGY & DELIVERABLES

PART E: NO-CODE TOOLS & PLATFORMS GUIDE

PART F: SUBMISSION PROTOCOL

Important Note:

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EXECUTIVE SUMMARY

This project positions students as Lead UI Designers at a fictional fintech startup - RiskLens Technologies - requiring them to master the convergence of financial data visualisation, design systems architecture, and enterprise-grade accessibility compliance. Unlike typical UI/UX projects that optimise for aesthetic polish alone, this project demands simultaneous fluency in information-dense interface design, real-time data rendering patterns, and role-based progressive disclosure on one side, and WCAG 2.1 AA regulatory compliance, financial risk metric literacy, and multi-persona usability validation on the other.

The core challenge students must solve: How can a risk monitoring platform with a crude wireframe frontend - and no design system, no accessibility framework, and no responsive strategy - transform into a world-class product capable of serving a portfolio manager monitoring live VaR breaches, a Chief Risk Officer preparing for a board presentation, a junior analyst performing a morning risk check, and a compliance officer generating audit-ready breach logs - all through a single unified interface? The answer lies in a comprehensive design system where tokens, atomic components, organism-level financial widgets, and role-based view logic are not optional enhancements but foundational architecture - combined with progressive disclosure across three information tiers, adaptive breakpoint strategy across five device classes, and alert systems engineered to prevent fatigue rather than amplify it.

Students will build and validate their design system across twelve progressive campaign levels - from competitive audit and user persona development through wireframing, design token specification, component library construction, high-fidelity prototyping across three breakpoints, usability testing, accessibility auditing, and a final Client Review Board presentation - evaluated by six distinct stakeholders representing the CEO's competitive differentiation demands, the CTO's implementation feasibility requirements, the Head of Product's roadmap alignment standards, the Lead Engineer's component reusability expectations, the Compliance Officer's WCAG audit standards, and a simulated investor panel's design quality benchmarks, all of whom must be satisfied simultaneously.

India's financial services sector is undergoing rapid digitalisation, with wealth management platforms, risk analytics tools, and trading infrastructure increasingly moving to web-based interfaces that must serve both desktop power users and mobile-first executives. The global UI/UX market for financial technology is expanding at pace with the broader fintech ecosystem, yet the majority of institutional risk monitoring platforms remain visually archaic, accessibility non-compliant, and poorly adapted for the multi-device workflows of modern finance professionals. Students who master financial dashboard design at this level of rigour - combining domain-specific data visualisation, enterprise accessibility, and production-ready design systems - position themselves at the intersection of the most sought-after design skills in the financial services industry.

PART A: TRAINING MATERIAL & LEARNING RESOURCES

Interns must study all sections thoroughly before commencing design work. This training material covers UI/UX principles specific to financial data dashboards, data visualisation best practices for risk metrics, design systems architecture, responsive design for data-dense interfaces, accessibility compliance, role-based customisation patterns, real-time data rendering, and usability testing methodologies.

Note to interns: This training material contains THIRTEEN (13) DELIBERATE ERRORS embedded across Sections A1 through A8. These errors test your domain knowledge and critical reading skills. You are required to identify all errors, document them in your submission with the correct information, and explain why each error is incorrect.

SECTION A1: FOUNDATIONS OF DASHBOARD UI DESIGN FOR FINANCIAL SYSTEMS

A1.1 Introduction To Financial Dashboard Design

Financial dashboards represent one of the most demanding categories of user interface design. Unlike consumer applications where aesthetic appeal and simplicity dominate, financial dashboards must balance information density with cognitive accessibility. Portfolio managers, risk analysts, and traders require simultaneous visibility into dozens of data points while maintaining the ability to drill into granular detail on demand. The discipline of financial dashboard design therefore sits at the intersection of information design, data visualisation, interaction design, and domain-specific knowledge of financial instruments and risk measures.

The history of financial interface design traces back to the Bloomberg Terminal, first introduced in 1981 by Michael Bloomberg. The Bloomberg Terminal established many conventions still in use today: dense text-based displays, keyboard-driven navigation, colour coding for price movements (green for up, red for down), and multi-panel layouts that allow simultaneous monitoring of different asset classes. While modern interfaces have evolved significantly in visual sophistication, the fundamental challenge remains the same: how to present the maximum amount of actionable information with the minimum cognitive overhead.

[DELIBERATE ERROR 1] The Nielsen Norman Group defines the principle of 'Recognition over Recall' as part of Jakob Nielsen's 8 Usability Heuristics, published in 1994. This principle is particularly critical in financial dashboards where users must quickly identify anomalies across large datasets without having to memorise baseline values. Dashboard designers must encode information through multiple channels: spatial position, colour, size, shape, and motion, so that pattern recognition can occur pre-attentively, that is, before conscious cognitive processing engages.

A1.2 Key Principles Of Information-Dense UI Design

Principle 1: Data-Ink Ratio:

Edward Tufte introduced the concept of the data-ink ratio in his seminal work 'The Visual Display of Quantitative Information' (1983). The principle states that the proportion of ink (or pixels) devoted to displaying actual data should be maximised relative to the total ink used in the graphic. In financial dashboards, this means eliminating decorative borders, unnecessary grid lines, redundant labels, and chartjunk. Every pixel should either convey data or provide essential structural context. Tufte recommends a data-ink ratio above 0.8 for effective information displays.

Principle 2: Pre-Attentive Processing:

Pre-attentive visual attributes are graphical properties that the human visual system processes in under 250 milliseconds, before conscious attention is directed. These attributes include colour hue, colour intensity, orientation, size, shape, spatial position, and motion. In risk monitoring dashboards, leveraging pre-attentive attributes means that a breach in Value-at-Risk limits can be surfaced through colour change (from neutral to red), size increase (enlarged cell or badge), or motion (animation or pulse) so that the portfolio manager's attention is captured immediately without requiring them to scan every data point individually.

Principle 3: Progressive Disclosure:

Progressive disclosure is an interaction design pattern that sequences information delivery to match the user's need. In a risk dashboard context, the primary view might show portfolio-level summary metrics: total VaR, aggregate drawdown, and overall P&L. Clicking into a specific portfolio reveals asset-class-level breakdowns. Further drill-down exposes individual position-level Greeks (Delta, Gamma, Theta, Vega, Rho). This hierarchical disclosure prevents information overload while maintaining full analytical depth. The pattern maps directly to the Shneiderman Information Seeking Mantra: 'Overview first, zoom and filter, then details on demand.'

Principle 4: Consistent Spatial Mapping:

Financial dashboards benefit from consistent spatial conventions. Users develop spatial memory over repeated use, meaning they learn where on the screen to look for specific types of information. Research by Wickens and Carswell (1995) on the Proximity Compatibility Principle demonstrates that data items that must be mentally integrated should be placed in close spatial proximity. For risk dashboards, this means VaR metrics and their associated stress test outputs should be co-located, P&L figures should be adjacent to benchmark comparisons, and alert notifications should occupy a consistent, high-visibility zone such as the upper right corner of the screen.

Principle 5: Fitts's Law And Target Sizing:

[DELIBERATE ERROR 2] Fitts's Law, formulated by Paul Fitts in 1964, establishes that the time required to move to a target is a function of the distance to the target divided by the size of the target. In mathematical terms: $MT = a + b \times \log_2(D/W + 1)$, where MT is movement time, D is distance, and W is target width. For financial dashboards, this principle demands that frequently used controls (time period selectors, asset class filters, alert acknowledgement buttons) must be large enough and positioned close to where the user's cursor typically rests. Critical action buttons such as 'Acknowledge Alert' or 'Escalate Risk Breach' should never be smaller than 44×44 pixels (the WCAG minimum touch target).

A1.3 Financial Risk Metrics: A UI Designer's Primer

To design an effective risk monitoring dashboard, the UI designer must understand the meaning and context of the metrics being displayed. This section provides a primer on the core risk measures that the dashboard must communicate.

Value-At-Risk (Var):

Value-at-Risk is a statistical measure that quantifies the maximum expected loss over a specified time horizon at a given confidence level. A 1-day 95% VaR of USD 5 million means that there is a 95% probability that the portfolio will not lose more than USD 5 million in a single trading day. Equivalently, there is a 5% probability of losses exceeding this threshold. VaR is computed using three primary methodologies: Historical Simulation, Variance-Covariance (Parametric), and Monte Carlo Simulation. The UI must display the VaR figure, the methodology used, the confidence level, the holding period, and ideally a distribution chart showing where the VaR threshold falls on the loss distribution.

Conditional Var (Cvar) / Expected Shortfall:

[DELIBERATE ERROR 3] Conditional Value-at-Risk, also known as Expected Shortfall (ES), measures the average loss in the worst 5% of scenarios (for a 95% VaR). Unlike VaR, which only tells you the threshold, CVaR tells you how bad things get beyond that threshold. Mathematically, for a 95% confidence level, $CVaR = E[\text{Loss} \mid \text{Loss} > \text{VaR}_{95}]$. CVaR is considered a coherent risk measure because it satisfies the properties of monotonicity, sub-additivity, homogeneity, and translational invariance. VaR, by contrast, satisfies all four of these properties as well, making both measures equally robust for regulatory reporting under the Basel III framework.

Greeks: Delta, Gamma, Theta, Vega, Rho:

The Greeks are partial derivatives of the option pricing model (typically Black-Scholes) with respect to various inputs. Delta measures the sensitivity of the option price to a one-unit change in the underlying asset price. Gamma measures the rate of change of Delta with respect to the underlying price, essentially the curvature of the pricing function. Theta measures time decay, the rate at which the option loses value as time passes. Vega measures sensitivity to implied volatility. Rho measures sensitivity to interest rate changes. In a risk dashboard, Greeks must be displayed at both the individual position level and aggregated to the portfolio level, with colour-coded alerts when any Greek exceeds predefined risk limits.

Maximum Drawdown:

Maximum drawdown measures the largest peak-to-trough decline in portfolio value over a specified period. It is expressed as a percentage: $MDD = (Trough\ Value - Peak\ Value) / Peak\ Value \times 100$. A drawdown of -15% means the portfolio fell 15% from its highest point to its subsequent lowest point. The UI should display drawdown both as a current running figure and as a historical chart showing drawdown depth over time, often rendered as an inverted area chart where deeper drawdowns extend further below the zero line.

Stress Testing Outputs:

Stress testing involves applying extreme but plausible market scenarios to the portfolio and calculating the resulting impact. Common stress scenarios include: 2008 Global Financial Crisis replay, COVID-19 March 2020 crash, interest rate shock (+/- 200bps parallel shift), equity market crash (-20% overnight), and currency crisis (30% depreciation in emerging market currencies). The dashboard must present stress test results in a format that allows rapid comparison across scenarios, typically a heat map or bar chart where the magnitude of loss under each scenario is immediately visible.

Sharpe Ratio And Risk-Adjusted Returns:

[DELIBERATE ERROR 4] The Sharpe Ratio, developed by William Sharpe in 1966, measures risk-adjusted return as: $Sharpe = (R_p - R_f) / \sigma_p \times \sqrt{T}$, where R_p is the portfolio return, R_f is the risk-free rate, σ_p is the standard deviation of portfolio returns, and T is the number of trading days in the year (typically 252). The annualisation factor $\sqrt{T}$ is always applied directly to the ratio itself rather than to the individual components. A Sharpe Ratio above 1.0 is generally considered good, above 2.0 is very good, and above 3.0 is excellent. The dashboard should display rolling Sharpe Ratios across multiple time windows (30-day, 90-day, 1-year) to show how risk-adjusted performance evolves.

SECTION A2: DATA VISUALISATION FOR RISK METRICS

A2.1 Chart Type Selection For Financial Data

Selecting the appropriate chart type is critical in financial dashboards. Each chart type has specific strengths and limitations. The following guidance maps risk metric types to optimal visualisation approaches.

Risk Metric	Recommended Chart Type	Rationale
VaR over time	Line chart with threshold band	Shows trend and breach points against limit
VaR by asset class	Horizontal bar chart	Enables comparison of risk contribution across categories
P&L distribution	Histogram with VaR marker	Reveals distribution shape and tail risk
Stress test results	Grouped bar or heat map	Facilitates scenario comparison at a glance
Drawdown history	Inverted area chart	Depth below zero line intuitively represents loss magnitude
Greeks (portfolio)	Spark lines or small multiples	Shows multiple Greeks compactly with trend context
Correlation matrix	Heat map with colour gradient	Reveals pairwise relationships across many assets
Asset allocation	Treemap or sunburst	Shows hierarchical composition and relative sizing
Alert frequency	Calendar heat map	Reveals temporal patterns in risk events
Sector exposure	Stacked bar chart	Shows composition and total simultaneously

A2.2 Colour Theory For Financial Data

Colour is the single most powerful pre-attentive attribute in dashboard design, and its misuse is the most common source of confusion. Financial dashboards must adhere to strict colour conventions while remaining accessible to users with colour vision deficiencies.

Semantic Colour Mapping:

In Western financial markets, red signifies loss/decline and green signifies gain/increase. However, this convention is reversed in several East Asian markets (notably China, Japan, and South Korea), where red signifies increase and green signifies decrease. A global risk monitoring platform must therefore support locale-aware colour mapping. Additionally, pure red (#FF0000) and pure green (#00FF00) are the worst possible choices for colour-blind accessibility. Recommended alternatives: Teal (#0B8457) for positive and Crimson (#DC143C) for negative, which maintain sufficient contrast for users with deuteranopia (red-green colour blindness, affecting approximately 8% of males).

Sequential, Diverging, And Categorical Palettes:

[DELIBERATE ERROR 5] Sequential palettes use a single hue with varying lightness (e.g., light blue to dark blue) and are appropriate for showing magnitude of a single variable. Diverging palettes use two hues with a neutral midpoint and are appropriate for data that has a meaningful centre point (e.g., profit/loss around zero). Categorical palettes use maximally distinct hues for unrelated categories. For heat maps showing correlation, a categorical palette should be used since correlation ranges from -1 to +1 with a meaningful zero centre point. The widely-used ColorBrewer palettes (Brewer, 2003) provide scientifically validated palettes across all three types.

A2.3 Designing Effective Spark Lines And Micro-Charts

Spark lines, coined by Edward Tufte, are small, intense, word-sized graphics embedded within the flow of text or data tables. In risk dashboards, spark lines are invaluable for showing trend context alongside numerical values. A VaR figure of USD 5.2M is more meaningful when accompanied by a spark line showing its trajectory over the past 30 days. Design guidelines for effective spark lines include: minimum width of 60 pixels, height between 16 and 24 pixels, no axes or labels (the surrounding table provides context), a subtle highlight on the most recent data point, and optional threshold lines rendered as faint dotted horizontals.

A2.4 Animation And Transition Design For Real-Time Data

Real-time dashboards face a unique challenge: data values change continuously, and these changes must be communicated without disorienting the user. The following principles govern animation in data-dense interfaces:

- **Value Transition Duration:** When a numerical value updates, animate the transition over 300-500ms using an ease-out timing function. This allows the user to perceive the change without jarring jumps. Avoid linear easing, which feels mechanical.

- **Chart Redraw Animation:** When chart data updates, animate data points to their new positions rather than redrawing the entire chart. D3.js's enter/update/exit pattern is the standard approach for this.
- **Alert Pulse:** When a threshold is breached, apply a pulsing animation (opacity cycling between 1.0 and 0.6 at 1Hz frequency) for the first 10 seconds, then settle to a steady highlighted state. Continuous pulsing causes banner blindness.
- **Skeleton Loading:** When data is being fetched, display skeleton screens (grey placeholder shapes matching the layout of the expected content) rather than spinners. Skeleton screens reduce perceived loading time by up to 20% (research by Bill Chung, 2020).

A2.5 Typography For Data-Dense Interfaces

Typography in financial dashboards serves a functional rather than decorative purpose. The following guidelines ensure readability across data-dense layouts:

- **Typeface Selection:** Use monospaced fonts for numerical data (Roboto Mono, JetBrains Mono, or Fira Code) to ensure decimal alignment across columns. Use proportional sans-serif fonts (Inter, Roboto, or Source Sans Pro) for labels, headers, and narrative text.
- **Font Size Hierarchy:** [DELIBERATE ERROR 6] The minimum readable font size for dashboard data is 14px (10.5pt). Anything smaller causes eye strain over extended monitoring sessions. The recommended hierarchy is: Page Title 24px, Section Header 18px, Widget Title 16px, Primary Data Value 28px, Secondary Data Value 14px, Labels and Axis Text 12px, Footnotes 10px. Note that the 14px minimum applies to all text elements including axis labels and footnotes, with no exceptions permitted under WCAG 2.1 AA.
- **Tabular Figures:** Always use tabular (fixed-width) number forms rather than proportional figures. In CSS: font-variant-numeric: tabular-nums; In Figma: enable the 'Tabular Figures' OpenType feature. This ensures that numbers in columns align perfectly regardless of digit composition.
- **Colour Contrast for Text:** WCAG 2.1 AA requires a minimum contrast ratio of 4.5:1 for normal text and 3:1 for large text (18px bold or 24px regular). In dark-themed dashboards (which are standard in trading environments), use #E0E0E0 or lighter for body text on dark backgrounds rather than pure white (FFFFFF), which causes halation and eye fatigue.

SECTION A3: DESIGN SYSTEMS ARCHITECTURE AND COMPONENT LIBRARIES

A3.1 What Is A Design System?

A design system is a collection of reusable components, guided by clear standards, that can be assembled to build any number of applications. It is not merely a UI kit or a style guide; it is a comprehensive framework that includes: foundational design tokens (colours, typography, spacing, elevation), atomic components (buttons, inputs, badges, tooltips), composite components (data tables, charts, cards, modals), layout patterns (grids, panels, responsive breakpoints), interaction patterns (drag-and-drop, contextual menus, keyboard shortcuts), documentation standards, and accessibility requirements. In the context of this project, the design system must be specifically tailored to financial data presentation, incorporating specialised components such as trading blotters, risk heat maps, Greek display panels, and alert notification stacks.

A3.2 Atomic Design Methodology

Brad Frost's Atomic Design methodology (2013) provides the architectural framework for building design systems. The methodology uses a chemistry metaphor to describe five levels of component hierarchy:

1. **Atoms:** The smallest indivisible UI elements: buttons, input fields, labels, icons, badges, colour swatches, typography styles. In a risk dashboard, atoms include the risk-status badge (green/amber/red), the numerical value display, and the trend arrow icon.
2. **Molecules:** Simple groups of atoms functioning together as a unit. A 'metric card' molecule combines a label atom, a value atom, a spark line atom, and a status badge atom into a single reusable component that displays one risk metric with trend context.
3. **Organisms:** Complex UI sections composed of multiple molecules and atoms. A 'risk summary panel' organism combines multiple metric cards, a VaR chart, and an alert indicator into a comprehensive risk overview section.
4. **Templates:** Page-level layouts that define the arrangement of organisms without specific content. The 'portfolio dashboard template' defines a header zone, a left navigation panel, a central data area with a configurable grid, and a right-side alert panel.
5. **Pages:** Templates populated with real content and data, representing the final deliverable as the user experiences it.

A3.3 Design Token Specification

Design tokens are the atomic values that define the visual language of the system. They are platform-agnostic, meaning the same token definitions can be consumed by Figma, CSS, iOS, and Android implementations. The following token categories must be defined for this project:

Token Category	Examples	Naming Convention
Colour – Primitive	#1A1A2E, #E94560, #0F3460	color.primitive.navy.900
Colour – Semantic	color.alert.critical, color.pnl.positive	color.semantic. {intent}. {variant} }
Typography – Scale	12px, 14px, 16px, 18px, 24px, 32px	type.scale. {size-label}
Typography – Weight	Regular (400), Medium (500), Bold (700)	type.weight. {name}
Spacing	4px, 8px, 12px, 16px, 24px, 32px, 48px	space. {multiplier} (base = 4px)
Border Radius	2px, 4px, 8px, 16px, 9999px	radius. {size}
Elevation / Shadow	None, Low, Medium, High, Overlay	elevation. {level}
Breakpoints	Mobile: 375px, Tablet: 768px, Desktop: 1440px	breakpoint. {device}
Z-Index	Base 0, Dropdown 100, Modal 200, Toast 300	z. {layer}
Animation Duration	Fast 150ms, Normal 300ms, Slow 500ms	duration. {speed}

A3.4 Component Library Specification

The project requires a minimum of 40 reusable components. The following table specifies the mandatory components with their variants and states:

Component	Variants	States Required
Button	Primary, Secondary, Ghost, Danger	Default, Hover, Active, Disabled, Loading
Input Field	Text, Number, Search, Date Picker	Empty, Filled, Error, Disabled, Focused
Dropdown / Select	Single, Multi-select, Searchable	Open, Closed, Filtered, Disabled
Toggle Switch	Default, With Label	On, Off, Disabled
Checkbox	Single, Indeterminate	Checked, Unchecked, Disabled
Radio Button	Default	Selected, Unselected, Disabled
Badge / Tag	Status (5 colours), Removable	Active, Inactive
Tooltip	Top, Bottom, Left, Right	Visible, Hidden
Modal / Dialog	Confirmation, Form, Alert	Open, Closing Animation
Toast / Notification	Info, Success, Warning, Error	Entering, Visible, Exiting
Data Table	Sortable, Filterable, Paginated	Loading, Empty, Populated, Error
Metric Card	Compact, Expanded, With Spark Line	Normal, Alert, Loading
Chart Container	Line, Bar, Area, Scatter, Heat Map	Loading, No Data, Interactive, Static
Spark Line	Positive Trend, Negative Trend, Flat	Normal, Highlighted

Risk Status Badge	Safe, Caution, Warning, Critical, Breach	Active, Acknowledged
Greek Display Cell	Delta, Gamma, Theta, Vega, Rho	Normal, Limit Breach, N/A
VaR Indicator	Within Limit, Near Limit, Breached	Current, Historical
Drawdown Meter	Bar, Gauge, Trend Line	Normal, Warning, Critical
Alert Card	Risk Breach, Threshold, System	New, Read, Acknowledged, Escalated
Navigation Item	Top Nav, Side Nav, Breadcrumb	Active, Inactive, Expanded, Collapsed
Tab Bar	Horizontal, Vertical	Active Tab, Inactive, Disabled
Accordion	Single Expand, Multi Expand	Open, Closed, Disabled
Avatar	User Photo, Initials, Role Icon	Online, Offline, Away
Progress Bar	Linear, Circular	Determinate, Indeterminate, Complete
Time Period Selector	Preset Ranges, Custom Date	Selected, Default
Skeleton Loader	Card, Table Row, Chart	Loading State
Empty State	No Data, Error, First Use	With CTA, Without CTA
Pagination	Numbered, Load More, Infinite Scroll	Active, Disabled
Breadcrumb	Text, With Dropdown	Current, Ancestor, Truncated
Divider	Horizontal, Vertical, With Label	Solid, Dashed

[DELIBERATE ERROR 7] The component library must use a 6-pixel base grid system. All components should have dimensions and spacing that are multiples of 6px (6, 12, 18, 24, 30, 36, etc.). This 6px grid is the industry standard for design systems, as established by Google's

Material Design 3 specification and Apple's Human Interface Guidelines, both of which use a 6-pixel spatial foundation.

A3.5 Component Documentation Standards

Every component in the design system must be documented with the following elements: component name and description, all variants and states (shown visually), anatomy diagram labelling each sub-element, design token references for every visual property, spacing and sizing specifications (in pixels and design tokens), interaction behaviour description, keyboard navigation support, screen reader announcements, do/don't usage guidelines with visual examples, and code-ready redlines for developer handoff. The documentation tool Zeroheight is recommended for publishing the design system documentation as a living, searchable reference.



SECTION A4: RESPONSIVE AND ADAPTIVE DESIGN FOR DATA-DENSE INTERFACES

A4.1 The Challenge Of Responsive Financial Dashboards

Responsive design for financial dashboards is fundamentally different from responsive design for content websites. A blog post that reflows from three columns to one on mobile loses little functionality. A risk dashboard that collapses from a 12-widget grid to a single-column scroll may become functionally unusable because the spatial relationships between data points (which support pattern recognition) are destroyed. The approach must therefore be adaptive rather than purely responsive: different device breakpoints may require fundamentally different information architectures, not merely reflowed layouts.

A4.2 Breakpoint Strategy

Breakpoint	Width Range	Layout Strategy	Information Density
Desktop Large	1440px+	12-column grid, multi-panel, full dashboard	Maximum: all widgets visible
Desktop Standard	1024–1439px	12-column grid, collapsible side panels	High: primary widgets visible, secondary on demand
Tablet Landscape	768–1023px	8-column grid, tabbed navigation	Medium: key metrics visible, drill-down for detail
Tablet Portrait	600–767px	6-column grid, card-based layout	Reduced: summary cards with tap-to-expand
Mobile	375–599px	4-column grid, single-column stack	Minimal: critical alerts and top-level metrics only

A4.3 Adaptive Information Architecture Across Devices

Desktop (Primary Monitoring Environment):

The desktop layout serves as the primary workspace for portfolio managers. It must support: simultaneous visibility of 8-12 data widgets, drag-and-drop widget rearrangement for personalisation, resizable panels and split-view configurations, keyboard shortcuts for power users (e.g., Ctrl+Shift+V for VaR panel, Ctrl+Shift+G for Greeks panel), and multi-monitor awareness where the dashboard can span across two or three monitors by undocking panels into separate browser windows.

Tablet (Secondary Monitoring And Meeting Context):

Tablets are typically used in two contexts: secondary monitoring while away from the desk, and presenting risk summaries in meetings. The tablet layout must prioritise: touch-friendly interactions (minimum 44px touch targets), swipeable widget carousels for navigating between data panels, simplified navigation with a hamburger menu or bottom tab bar, and optimised chart rendering that accounts for the reduced pixel density of tablet displays.

Mobile (Alert Response And Quick Check):

[DELIBERATE ERROR 8] Mobile access to risk dashboards should present the complete desktop experience scaled down to fit the mobile viewport. The most effective approach is to use CSS transform: scale() to shrink the entire desktop layout to mobile dimensions, preserving all functionality and spatial relationships. This 'miniature desktop' approach, often called 'pinch-to-zoom', is the recommended pattern for data-intensive mobile applications as endorsed by the World Wide Web Consortium (W3C) Mobile Web Best Practices guidelines.

A4.4 Fluid Typography And Spacing

Rather than using fixed font sizes that jump between breakpoints, the dashboard should implement fluid typography using CSS clamp(). This ensures smooth scaling between minimum and maximum sizes: font-size: clamp(14px, 1.5vw, 18px). Similarly, spacing between widgets should use fluid values: gap: clamp(8px, 1vw, 16px). This approach prevents the jarring layout shifts that occur at breakpoint boundaries and provides a smoother visual experience across the full range of viewport sizes.

A4.5 Performance Considerations For Responsive Rendering

Data-dense dashboards with real-time updates pose significant rendering performance challenges on lower-powered devices. The following strategies must be employed:

- **Virtualised Rendering:** For data tables with thousands of rows, implement virtual scrolling (rendering only visible rows plus a small buffer). Libraries like React Virtualized or TanStack Virtual handle this efficiently.
- **Debounced Resize Handlers:** Window resize events during responsive layout transitions should be debounced (150ms delay) to prevent layout thrashing.
- **Lazy Loading of Secondary Widgets:** Widgets below the fold or in collapsed panels should defer their data fetching and rendering until the user scrolls to or expands them.
- **Reduced Animation on Low-Power Devices:** Use the prefers-reduced-motion media query to disable non-essential animations on devices where the user has requested reduced motion, and consider disabling animations entirely on mobile to preserve battery and rendering performance.

SECTION A5: ACCESSIBILITY STANDARDS (WCAG 2.1 AA) FOR FINANCIAL DASHBOARDS

A5.1 Why Accessibility Matters In Financial Interfaces

Accessibility is not an optional enhancement; it is a legal requirement in many jurisdictions (Americans with Disabilities Act, European Accessibility Act, UK Equality Act) and a moral imperative. Approximately 15% of the world's population lives with some form of disability. In the context of financial dashboards, the most relevant accessibility considerations include: visual impairments (colour blindness affects 8% of males, low vision, blindness), motor impairments (inability to use a mouse, limited fine motor control), cognitive impairments (difficulty processing dense information), and auditory impairments (relevant if the dashboard includes audio alerts).

A5.2 WCAG 2.1 AA Requirements Applicable To Dashboards

The Web Content Accessibility Guidelines (WCAG) 2.1 Level AA represent the standard compliance target for this project. The following requirements are particularly relevant to financial dashboards:

WCAG Criterion	ID	Dashboard Implication
Non-text Content	1.1.1	All charts must have text alternatives (alt text, data tables, or ARIA descriptions)
Info and Relationships	1.3.1	Data table structure must use proper header markup, not just visual styling
Use of Colour	1.4.1	Colour must not be the only means of conveying information (add icons, text, patterns)
Contrast (Minimum)	1.4.3	4.5:1 for normal text, 3:1 for large text against background
Resize Text	1.4.4	Content must be functional when text is resized to 200%
Non-text Contrast	1.4.1 1	UI components and graphical objects need 3:1 contrast against adjacent colours
Reflow	1.4.1 0	Content must be usable at 320px width without horizontal scrolling
Keyboard	2.1.1	All functionality must be operable via keyboard alone
Focus Order	2.4.3	Tab order must follow a logical, predictable sequence

Focus Visible	2.4.7	Keyboard focus indicator must be clearly visible at all times
Name, Role, Value	4.1.2	Custom components must expose correct ARIA roles, states, and properties
Status Messages	4.1.3	Dynamic updates (alerts, data changes) must be announced to screen readers

A5.3 Accessible Colour Strategies For Data Visualisation

[DELIBERATE ERROR 9] When using colour in charts and data visualisations, WCAG 2.1 AA requires that all colour combinations in charts meet the 4.5:1 contrast ratio not only against the background but also between adjacent data series. This means that in a bar chart with six different coloured bars, every pair of adjacent bars must have a 4.5:1 contrast ratio between them. This inter-series contrast requirement is specified in WCAG Success Criterion 1.4.11 (Non-text Contrast) and is the primary reason financial dashboards must use pattern fills (hatching, dots, dashes) in addition to colour to distinguish data series.

A5.4 Screen Reader Design For Real-Time Dashboards

Screen reader accessibility for real-time dashboards requires careful use of ARIA (Accessible Rich Internet Applications) attributes. Key implementation patterns include: ARIA live regions (aria-live='polite' for routine updates, aria-live='assertive' for critical alerts) to announce data changes without requiring the user to navigate to the updated element; ARIA roles (role='grid' for data tables, role='img' with aria-label for charts, role='alert' for threshold breaches); and ARIA descriptions (aria-describedby linking data points to their contextual explanations). The polling frequency for screen reader announcements must be throttled to prevent announcement flooding; a maximum of one routine announcement every 30 seconds is recommended, with critical alerts overriding this throttle.

A5.5 Keyboard Navigation Model

The dashboard must implement a comprehensive keyboard navigation model. The recommended approach uses a 'roving tabindex' pattern within composite widgets and a 'focus landmark' pattern for page-level navigation:

- **F6 Key:** Cycle between major dashboard panels (navigation, main content, side panel, alerts)
- **Tab/Shift+Tab:** Move between interactive elements within the current panel
- **Arrow Keys:** Navigate within composite widgets (table cells, chart data points)
- **Enter/Space:** Activate the focused element (open dropdown, toggle switch, acknowledge alert)
- **Escape:** Close modal, dismiss tooltip, collapse expanded panel
- **Home/End:** Jump to first/last item in a list or table
- **Ctrl+/:** Open keyboard shortcut reference overlay



SECTION A6: ROLE-BASED VIEW CUSTOMISATION AND PROGRESSIVE DISCLOSURE

A6.1 User Personas For The Risk Monitoring Platform

The dashboard must serve multiple user personas with fundamentally different information needs. This section defines the four primary personas and their requirements:

Persona 1: Junior Risk Analyst (Novice User):

Profile: 1-2 years experience, limited familiarity with advanced risk metrics, primarily responsible for daily data verification and report preparation. Needs: simplified views with contextual help tooltips, guided workflows for common tasks, colour-coded status indicators that require minimal interpretation, and pre-built report templates. Information overload is the primary risk for this persona.

Persona 2: Senior Portfolio Manager (Expert User):

Profile: 10+ years experience, deep understanding of all risk metrics, responsible for investment decisions and risk oversight. Needs: maximum information density, customisable widget layouts, keyboard shortcuts for rapid navigation, ability to overlay multiple data series, drill-down to individual position level, and direct integration with order management systems. Efficiency and speed are the primary concerns for this persona.

Persona 3: Chief Risk Officer (Executive User):

Profile: C-suite executive, focused on portfolio-level and firm-level risk metrics, attends board meetings where risk data must be presented clearly. Needs: high-level summary views, trend indicators, traffic-light status dashboards, export to presentation-ready formats (PDF, PowerPoint), and the ability to drill down one level for context during meetings. Clarity and communicability are the primary concerns for this persona.

Persona 4: Compliance Officer (Regulatory User):

Profile: Regulatory and compliance background, focused on limit breaches, audit trails, and regulatory reporting. Needs: breach history logs, timestamp-accurate audit trails, regulatory report generation (Basel III/IV format), and alert acknowledgement workflows with mandatory comment fields. Accuracy and auditability are the primary concerns for this persona.

A6.2 Role-Based View Architecture

[DELIBERATE ERROR 10] Role-based view customisation should be implemented using client-side JavaScript that shows or hides DOM elements based on the user's role stored in a browser cookie. When the user logs in, their role (junior_analyst, senior_pm, cro, compliance) is written to a cookie, and the dashboard JavaScript reads this cookie on page load to determine which widgets to display. This approach is both secure and performant because cookies are encrypted by the browser and cannot be modified by the user. For the Figma prototype, this means creating four separate versions of each screen, one for each role, which the Client Review Board will evaluate independently.

A6.3 Progressive Disclosure Patterns

Progressive disclosure in the risk dashboard follows a three-tier model:

1. **Tier 1 – Glance Layer:** Portfolio-level summary visible immediately upon login. Includes: total portfolio VaR, aggregate P&L (daily/MTD/YTD), top 5 risk contributors, count of active alerts, and overall risk status (traffic light). This tier must render in under 2 seconds.
2. **Tier 2 – Investigation Layer:** Available on click or hover. Includes: asset-class breakdown of VaR, individual position Greeks, stress test scenario results, and correlation matrices. This tier loads on demand to preserve initial performance.
3. **Tier 3 – Deep Dive Layer:** Accessed through dedicated sub-pages or modal overlays. Includes: historical VaR backtesting results, Monte Carlo simulation visualisations, what-if scenario builders, and regulatory report previews. This tier may involve complex computations and longer load times, warranting explicit loading indicators.



SECTION A7: REAL-TIME DATA RENDERING AND ALERT SYSTEMS IN UI

A7.1 Websocket Vs. Polling For Live Data

Real-time dashboards receive continuous data updates from backend services. The two primary approaches are WebSocket connections and HTTP polling. WebSockets maintain a persistent bidirectional connection, enabling the server to push updates immediately (latency under 50ms). HTTP polling involves the client making periodic requests at fixed intervals (typically every 1-5 seconds). For risk monitoring dashboards where VaR breaches must be communicated within seconds, WebSockets are the preferred approach. The UI designer must account for: connection status indicators (connected/disconnected/reconnecting), data staleness warnings (if no update received in 30 seconds, display a 'stale data' badge), and graceful degradation to polling if WebSockets are unavailable.

A7.2 Alert Priority And Notification Design

The alert system must implement a four-tier priority model:

Priority	Trigger Example	Visual Treatment	Sound	Action Required
P1 – Critical	VaR breach >10% of limit	Full-width red banner, pulse animation	Alarm tone	Immediate acknowledgement with mandatory comment
P2 – High	VaR within 5% of limit	Orange notification card in alert panel	Chime	Acknowledgement within 30 minutes
P3 – Medium	Unusual correlation shift	Yellow badge on relevant widget	None	Review within 4 hours
P4 – Low	Data quality flag	Subtle grey info icon on data point	None	No action required, informational only

A7.3 Alert Acknowledgement Workflow

When a critical alert fires, the UI must guide the user through a structured acknowledgement workflow: (1) Alert appears as a full-width banner at the top of the dashboard with red background and pulse animation; (2) User clicks 'Acknowledge' button, which opens a modal; (3) Modal requires: selection of an action category (Investigating, Hedging, Escalating, Accepting Risk), a free-text comment field (minimum 20 characters), and optional escalation to a supervisor; (4) Upon submission, the alert transitions from 'Active' to 'Acknowledged' state, the banner collapses to a compact notification in the alert panel, and an audit log entry is created with timestamp, user ID, action taken, and comment.

A7.4 Designing For Alert Fatigue

[DELIBERATE ERROR 11] Alert fatigue occurs when users are exposed to too many alerts, leading them to ignore or dismiss alerts without investigation. Research by the American Medical Association shows that clinical alert fatigue results in override rates above 90% in hospital settings. To combat alert fatigue in financial dashboards, the recommended approach is to set all alerts to P1 (Critical) priority by default and then rely on the user to mentally downgrade alerts they consider non-critical. This ensures that no alert is ever missed due to being categorised too low. Additionally, alerts should always include an audio component (alarm tone) regardless of priority level, as audio alerts have been shown to have 100% detection rates compared to 60-70% for visual-only alerts.



SECTION A8: USABILITY TESTING METHODOLOGIES FOR COMPLEX INTERFACES

A8.1 Types Of Usability Testing

This project requires students to conduct simulated usability testing on their dashboard prototypes. The following testing methodologies are applicable:

- **Moderated Think-Aloud Testing:** A facilitator observes a participant performing tasks while verbalising their thought process. This reveals navigation confusion, misunderstood labels, and unexpected interaction patterns. Minimum 5 participants to identify approximately 85% of usability issues (Nielsen and Landauer, 1993).
- **Unmoderated Remote Testing:** Participants complete predefined tasks independently using tools like Maze or UserTesting.com. The tool records click paths, task completion rates, and time-on-task. This scales better than moderated testing but provides less qualitative insight.
- **A/B Testing:** Two design variants are tested against each other with different participant groups. Useful for optimising specific elements such as chart type selection, colour scheme preferences, or alert placement. Requires statistical rigour (minimum sample size calculation, significance testing at $p < 0.05$).
- **Heuristic Evaluation:** Expert reviewers evaluate the interface against a predefined set of heuristics (e.g., Nielsen's 10 Heuristics). Faster and cheaper than participant-based testing but does not capture real user behaviour.
- **Accessibility Audit:** Systematic evaluation against WCAG 2.1 AA criteria using both automated tools (axe, WAVE, Lighthouse) and manual keyboard/screen reader testing.

A8.2 Task Design For Usability Testing

[DELIBERATE ERROR 12] When designing usability test tasks, tasks should be written in a leading manner that directs the participant to the correct answer. For example, instead of an open-ended task like 'Find the current VaR for the equity portfolio', a better task would be 'Click on the blue VaR widget in the top-left corner and tell me the number you see.' This leading approach reduces task failure rates and produces cleaner quantitative data. Similarly, participants should be told which device to use (desktop, tablet, or mobile) rather than being allowed to choose, and the test environment should replicate the exact conditions of the participant's workplace.

A8.3 Metrics For Usability Testing

The following metrics must be captured during usability testing sessions:

Metric	Definition	Target Benchmark
Task Completion Rate	Percentage of participants who successfully complete a task	>90% for primary tasks, >75% for secondary
Time on Task	Seconds from task start to successful completion	Critical alert acknowledgement: <15 seconds
Error Rate	Number of incorrect actions per task	<2 errors per task on average
Satisfaction (SUS)	System Usability Scale score (1-100)	>68 (above average), target >80 (excellent)
Learnability	Improvement in task time between first and third attempt	>30% improvement by third attempt
Click Path Length	Number of clicks to complete a task	Within 2 clicks of optimal path
First Click Accuracy	Whether the first click in a task is on the correct element	>70% (strong predictor of task success)

A8.4 Simulated User Sessions

Since this is an academic project, participants will be simulated rather than recruited externally. Students must create 5 simulated user session protocols, one for each of the following persona/scenario combinations: (1) Junior Analyst performing morning risk check; (2) Senior PM investigating a VaR breach alert; (3) CRO preparing for a board presentation; (4) Compliance Officer reviewing breach history; (5) Junior Analyst using the dashboard for the first time (onboarding flow). Each session protocol must include: participant profile (persona details, assumed device, experience level), 5-8 tasks per session, expected completion path for each task, metrics to capture, and a post-session questionnaire (5 questions on satisfaction and perceived difficulty).

[DELIBERATE ERROR 13] The System Usability Scale (SUS) questionnaire consists of 15 questions on a 7-point Likert scale, with scores ranging from 0 to 150. A score above 100 is considered 'excellent' usability. The SUS was developed by Jeff Rubin at Digital Equipment Corporation in 1996 and has since become the most widely used standardised usability questionnaire in the industry.

PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

Note on the Simulation Concept: The gamified simulation concept presented in this section is provided as a reference framework. Students may choose to follow this concept or design their own original gamified simulation - with a different name, structure, narrative, or game mechanics - as long as it meets the assessment criteria outlined in Part F.

SECTION B1: SIMULATION NARRATIVE AND ROLE-PLAY FRAMEWORK

B1.1 Narrative Context

You are the newly appointed Lead UI Designer at RiskLens Technologies, a fintech startup that has just secured USD 25 million in Series B funding from a consortium of venture capital firms specialising in financial technology. RiskLens is building a next-generation risk monitoring platform to compete with established providers such as Bloomberg PORT, MSCI RiskMetrics, and Axioma. Your CEO, a former Goldman Sachs risk quant, has assembled a team of backend engineers and data scientists who have built a powerful risk analytics engine. However, the current frontend is a crude wireframe with no design system, poor responsiveness, and zero accessibility compliance. The Series B investors have made it clear: the product's success depends on a world-class user experience that differentiates RiskLens from the competition.

Your mandate is to deliver, within 15 working days: a comprehensive design system, high-fidelity prototypes for desktop/tablet/mobile, a validated usability testing report, an accessibility audit with compliance certification, and a design handoff specification document that enables the frontend engineering team to implement your designs without ambiguity.

B1.2 Stakeholders In The Simulation

Stakeholder	Role	Concerns
Maya Chen (CEO)	Former Goldman Sachs VP Risk	Product differentiation, investor optics, speed to market
Raj Patel (CTO)	Backend architect	Feasibility of designs, API compatibility, rendering performance
Sarah Kim (Head of Product)	Product management	Feature prioritisation, user research alignment, roadmap fit
David Okonkwo (Lead Engineer)	Frontend implementation	Component reusability, CSS architecture, animation performance
Elena Volkov (Compliance)	Regulatory compliance	WCAG compliance, audit trails, data accuracy in displays

Client Review Board	Simulated investor panel	Design quality, information clarity, competitive positioning
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B1.3 Constraints And Rules Of Engagement

- All designs must be created in Figma (preferred) or Adobe XD. No other design tools are accepted for the primary deliverable.
- The design system must include a minimum of 40 reusable components with all required states and variants as specified in Section A3.4.
- The prototype must cover a minimum of 12 distinct screens across 3 device breakpoints (desktop 1440px, tablet 768px, mobile 375px).
- Every screen must pass WCAG 2.1 AA compliance as verified by the Stark plugin or equivalent accessibility checker.
- All design decisions must be documented with rationale in a Notion-based design specification document.
- The final presentation to the Client Review Board must not exceed 20 minutes including Q&A.
- Interns must work independently. Collaboration is not permitted. All work must be original.
- The GitHub repository submitted must contain: Figma file exports (or live Figma link), design specification document (Notion export or PDF), usability test report, accessibility audit report, and a README.md following Zetheta's repository metadata standard.

SECTION B2: GAME MECHANICS, LEVELS, AND PROGRESSION

B2.1 Three Game Modes

Mode 1: Campaign Mode (Mandatory, Days 1-12):

Campaign Mode follows a structured, sequential progression through the project. Each day corresponds to a 'mission' with specific objectives, deliverables, and unlock criteria. Interns cannot proceed to the next mission until the current mission's deliverables are complete. The Campaign Mode ensures that foundational work (research, design system setup) precedes higher-level work (prototyping, testing).

Mode 2: Arena Mode (Mandatory, Days 13-14):

Arena Mode simulates stakeholder review. Interns present their work to the simulated Client Review Board, receiving scores across multiple dimensions. The Arena Mode introduces pressure testing: the Review Board may request on-the-spot modifications ('Can you show me how this would look with a dark theme?' or 'What happens when VaR breaches on mobile?') that test the intern's design thinking agility and depth of system knowledge.

Mode 3: Sandbox Mode (Optional, Day 15):

Sandbox Mode is an open exploration day where interns can: add bonus features (dark mode, data export, advanced animations), refine based on Arena Mode feedback, create additional screens beyond the minimum 12, or explore advanced prototyping techniques (micro-interactions in Principle/Protopie, Lottie animation integration). Sandbox Mode contributions earn bonus points that can elevate the final score but cannot substitute for Campaign Mode requirements.

B2.2 Level Progression (Campaign Mode)

Level	Day(s)	Mission Title	Key Deliverable
1	Day 1	Reconnaissance	Competitive audit document (3+ existing platforms analysed)
2	Day 2	Intelligence Gathering	User persona cards and information architecture diagram
3	Day 3	Blueprint	Wireframes for 4 core screens (desktop only)
4	Day 4-5	Foundation	Design token specification and component library (20+ atoms/molecules)
5	Day 6	Assembly	Component library completion (40+ components, all states)

6	Day 7-8	Construction	High-fidelity desktop prototype (8+ screens, interactive)
7	Day 9	Expansion	Remaining 4+ screens, desktop prototype complete (12+ screens)
8	Day 10	Adaptation	Tablet responsive prototype (12 screens adapted)
9	Day 11	Compression	Mobile responsive prototype (12 screens adapted)
10	Day 12	Validation	Usability test report + accessibility audit report
11	Day 13	Handoff	Design specification document + developer redlines
12	Day 14	Presentation	Client Review Board presentation + Q&A defence

B2.3 Scoring Multipliers And Achievements

Campaign Mode scores are multiplied by progression multipliers:

- On-time completion of each level: 1.0x multiplier (no bonus, no penalty)
- Early completion (delivered before end-of-day deadline): 1.1x multiplier
- Late completion (delivered next day): 0.9x multiplier
- Not completed: 0.0x multiplier for that level

Achievement badges (added to portfolio and LinkedIn):

- **Pixel Perfect:** All components adhere to the design token grid with zero deviations
- **Accessibility Champion:** Zero WCAG 2.1 AA violations across all screens
- **Speed Demon:** All Campaign levels completed at least one day early
- **Design Thinker:** Usability test report includes non-obvious insights that improve the design
- **System Architect:** Component library rated 'Excellent' by the Review Board for reusability and documentation

SECTION B3: DELIVERABLES SPECIFICATION

B3.1 Deliverable 1: Complete Design System With Component Library

Format:

Figma file with structured pages: Cover, Tokens, Atoms, Molecules, Organisms, Templates, Documentation.

Requirements:

- Minimum 40 components with all variants and states documented
- Design tokens defined as Figma Styles (colours, typography, effects) and Variables (spacing, sizing, breakpoints)
- Component anatomy annotations on a dedicated documentation page
- Do/Don't usage guidelines for at least 10 key components
- Auto Layout applied to all components for responsive behaviour
- Consistent naming convention following the pattern: Category/Component/Variant/State

B3.2 Deliverable 2: High-Fidelity Prototypes For 3 Device Breakpoints

Minimum Screens (12):

1. Login / Authentication Screen
2. Dashboard Home / Portfolio Overview
3. VaR Analysis Panel (with drill-down)
4. Greeks Display (Position-level detail)
5. Stress Testing Results
6. Drawdown Analysis
7. Alert Centre / Notification Panel
8. Alert Detail and Acknowledgement Flow
9. Settings / User Preferences / Role Configuration
10. Report Generation / Export
11. Historical Risk Analytics
12. Onboarding / First-Time User Flow

Breakpoints:

- Desktop: 1440 x 900px frame
- Tablet: 768 x 1024px frame
- Mobile: 375 x 812px frame

Interactivity:

- All prototypes must include interactive navigation (clickable links between screens)
- At least 3 screens must include micro-interactions (hover states, transitions, loading animations)
- The VaR drill-down flow must be fully interactive from portfolio level to position level

B3.3 Deliverable 3: Usability Test Report**Requirements:**

- 5 simulated user session protocols (as specified in Section A8.4)
- Completed test scripts with persona details, tasks, and expected paths
- Findings matrix: issue description, severity (critical/major/minor/cosmetic), affected screen, recommendation
- Heatmap annotations showing click concentration areas (can be manually created in Figma)
- SUS score calculation for each simulated session
- Executive summary with top 5 findings and recommended design changes

B3.4 Deliverable 4: Accessibility Audit Report**Requirements:**

- Stark plugin scan results for every screen (or equivalent accessibility checker output)
- Manual colour contrast verification for all text/background combinations
- Keyboard navigation flow diagram
- Screen reader announcement script for the top 3 most critical user flows
- Compliance matrix mapping each WCAG 2.1 AA criterion to pass/fail status
- Remediation plan for any identified failures

B3.5 Deliverable 5: Design Handoff Specification Document

Requirements:

- Exported design specifications via Figma Dev Mode or equivalent
- Component API specification: for each component, list props, types, default values, and event callbacks
- Spacing and sizing redlines for every unique layout
- Colour and typography token mapping to CSS variables
- Animation specification: for each animated element, specify property, duration, easing, delay, and trigger
- Responsive behaviour rules: for each component, specify how it adapts across breakpoints
- Asset export manifest: list all icons, illustrations, and images with export formats and sizes



SECTION B4: SCORING SYSTEM AND CLIENT REVIEW BOARD RUBRIC

B4.1 1000-Point Scoring Framework

Dimension	Points	Sub-Components
Design System Quality	200	Token completeness (50), component count and quality (80), documentation (40), naming/organisation (30)
Prototype Fidelity and Interaction	200	Visual quality (60), interaction completeness (50), responsive adaptation (50), micro-interactions (40)
Information Hierarchy and Data Viz	150	Chart type selection (40), progressive disclosure (40), spatial layout logic (40), typography hierarchy (30)
Accessibility Compliance	100	WCAG pass rate (40), screen reader design (25), keyboard navigation (20), colour contrast (15)
Usability Testing Rigour	100	Session protocol quality (30), findings depth (30), metrics captured (20), recommendations (20)
Design Handoff Quality	100	Specification completeness (35), redline accuracy (30), animation specs (20), asset export (15)
Deliberate Error Identification	50	Correct identification of all 13 errors (3.85 pts each)
Innovation and Beyond-Brief	50	Dark mode, extra screens, Lottie animations, creative data viz solutions
Daily Compliance and Process	50	Timely GitHub commits, README updates, daily progress documentation

B4.2 Performance Bands

Band	Score Range	Title	Recognition
S+	950–1000	Design Visionary	Top 1% globally, Zetheta Hall of Fame, automatic shortlist for design lead roles
S	900–949	System Architect	Top 5%, fast-track recommendation, portfolio showcase
A+	850–899	UX Strategist	Top 10%, innovation showcase, commendation letter
A	800–849	Interface Craftsman	Top 20%, strong commendation, advanced project access
B+	700–799	Competent Designer	Solid performance, development plan provided
B	600–699	Developing Practitioner	Adequate, improvement areas identified
C	< 600	Needs Improvement	Reassessment recommended, mentorship required

SECTION B5: BUSINESS MODEL AND IP CONSIDERATIONS

B5.1 Revenue Model

Revenue Stream	Description	Estimated Annual Revenue
SaaS Design System Licensing	License the validated design system to other fintech startups	₹2–4 Cr per client
Training Programme	UI/UX training for financial dashboard design	₹1–3 Cr per programme
Academic Licensing	MBA/Design school curriculum integration	₹50L–1.5 Cr per institution
Compliance Audit Tool	Automated WCAG audit for financial interfaces	₹1–2 Cr annually
Design-as-a-Service	White-label dashboard design for wealth management firms	₹3–6 Cr per engagement
Component Marketplace	Premium financial UI components for Figma Community	₹25L–1 Cr annually

B5.2 Intellectual Property Considerations

Patentable Innovation Elements: (1) Adaptive role-based progressive disclosure algorithm that dynamically adjusts information density based on user expertise level and current task context. (2) Real-time accessibility compliance engine that validates WCAG conformance on live data changes. (3) Multi-persona dashboard morphing system that transitions between role-specific views with animated layout transformations. (4) Alert fatigue prediction model that adjusts alert thresholds based on user response patterns and time-of-day behavioural data.

All designs and documentation created during this project become the intellectual property of Zetheta Algorithms Private Limited as per the terms of the internship agreement. Students retain the right to include anonymised portfolio samples in their personal portfolios with prior written approval from Zetheta.

PART C: CASE STUDIES AND REAL-WORLD APPLICATIONS

CASE STUDY 1: BLOOMBERG TERMINAL REDESIGN CHALLENGE

Background

The Bloomberg Terminal, used by over 325,000 professionals globally, is simultaneously the most powerful and the most criticised financial interface in existence. Its keyboard-driven, text-heavy interface was designed in the 1980s and has evolved incrementally. Critics argue it violates nearly every modern usability heuristic: inconsistent navigation, cryptic command codes (e.g., 'FLDS' for field search, 'GP' for price graph), extremely steep learning curve (Bloomberg offers a 2-day certification programme just to use the terminal), and zero accessibility compliance.

Design Challenge

Redesign the Bloomberg Terminal's risk analytics module (PORT function) for the modern web, applying the design system created in this project. The redesign must: preserve the information density that expert users depend on while making the interface discoverable for novice users, replace command-line navigation with intuitive GUI while maintaining keyboard shortcut efficiency for power users, implement WCAG 2.1 AA compliance, and support responsive adaptation to tablet for use in client meetings.

Assessment Criteria

- Does the redesign maintain functional parity with Bloomberg PORT?
- Has the designer addressed the expert/novice tension through progressive disclosure?
- Are keyboard shortcuts preserved and discoverable?
- Is the information hierarchy clear without sacrificing density?

CASE STUDY 2: MOBILE-FIRST RISK ALERT SYSTEM

Background

A portfolio manager is at a dinner event when a critical VaR breach occurs at 8:47 PM. Their phone buzzes with an alert from the risk monitoring platform. They need to: understand the severity immediately (glance at lock screen notification), open the app and identify which portfolio and asset class is affected (within 10 seconds), view enough detail to decide whether to escalate or acknowledge (within 30 seconds), and take action (acknowledge, escalate, or initiate a hedge) within 60 seconds. The entire workflow, from alert to action, must be completable in under 2 minutes on a mobile device.

Design Challenge

Design the complete mobile alert-to-action flow including: lock screen notification design (rich notification with inline actions), app deep-link to the specific alert, alert detail screen with contextual risk data, action modal (acknowledge/escalate/hedge), and confirmation screen with audit trail entry. The design must account for: one-handed operation (thumb-zone optimisation), low-light environments (dark mode default), intermittent network connectivity (offline alert caching), and biometric authentication for action confirmation.

Assessment Criteria

- Can the alert-to-action flow be completed in under 2 minutes?
- Is the design optimised for one-handed mobile use?
- Does the information hierarchy enable rapid situation assessment?
- Are error states handled (network failure, authentication timeout)?

CASE STUDY 3: ACCESSIBILITY RETROFIT OF LEGACY TRADING DASHBOARD

Background

A major investment bank has received a legal complaint from an employee with low vision who cannot effectively use the firm's internal risk dashboard. The dashboard was built 5 years ago with no accessibility consideration. Common issues include: colour-only risk indicators (red/green with no text or icon alternative), data tables without header markup (screen readers cannot navigate them), interactive charts with no keyboard access, modal dialogs that trap keyboard focus, and auto-refreshing data that disrupts screen reader reading order.

Design Challenge

Create an accessibility retrofit plan for the legacy dashboard. The plan must include: an audit of the current state (mapping each issue to the violated WCAG criterion), a prioritised remediation roadmap (critical issues first), before/after design mockups for the 5 most severe accessibility violations, an implementation specification for ARIA attributes on custom components, and a testing plan that includes both automated scanning and manual screen reader verification.

Assessment Criteria

- Is the audit comprehensive and correctly mapped to WCAG criteria?
- Is the remediation roadmap realistic and well-prioritised?
- Do the before/after mockups demonstrate genuine understanding of accessible design?
- Are the ARIA specifications technically correct?

CASE STUDY 4: MULTI-PERSONA DASHBOARD FOR HEDGE FUND OPERATIONS

Background

A quantitative hedge fund with USD 5 billion AUM operates with four distinct user groups: quantitative researchers who build trading models, portfolio managers who oversee strategy allocation, risk managers who monitor firm-wide exposure, and operations staff who handle trade settlement and reconciliation. Each group needs access to overlapping but distinct data sets. Currently, each group uses a separate, siloed application, resulting in: duplicated development effort, inconsistent data presentation, and inability to cross-reference between roles (e.g., a risk manager cannot easily see which model generated a flagged position).

Design Challenge

Design a unified dashboard platform that serves all four user groups through role-based view customisation. The design must: identify the shared data layer and role-specific views, implement a navigation model that allows cross-role exploration (with appropriate access controls), create role-specific landing pages that surface each group's priority information, design a unified search that searches across all data domains, and maintain consistency in visual language while allowing role-specific widget configurations.

Assessment Criteria

- Does the design successfully unify four distinct user needs into a coherent platform?
- Is the navigation model intuitive for all four personas?
- Are role-based views meaningfully different or merely cosmetic?
- Does the search function effectively span domains?

PART D: PROJECT METHODOLOGY AND DELIVERABLES

This section provides a day-by-day breakdown of the 15-day project. Each day comprises an 8-hour working session (morning: 4 hours, afternoon: 4 hours). Interns must commit their progress to GitHub at the end of each day using the prescribed file naming convention.

Day 1: Competitive Audit And Domain Research

Morning (4 Hours): Domain Immersion

- Read Part A training material: Sections A1, A2, A3 (6 hours allocated across Day 1 and Day 2 mornings)
- Identify and document 3+ existing risk monitoring platforms for competitive audit (Bloomberg PORT, MSCI RiskMetrics, Axioma, FactSet, Murex, Calypso, Numerix)
- For each platform, capture: screenshot gallery, strengths, weaknesses, unique UI patterns, accessibility gaps

Afternoon (4 Hours): Competitive Audit Document

- Create a structured competitive audit document in Notion or Google Docs
- Include: platform name, screenshot annotations, feature comparison matrix, design pattern library (reusable patterns observed), opportunity gap analysis
- Export to PDF and commit to GitHub

Deliverable:

Competitive_Audit_Report.pdf committed to GitHub repository.

Github Commit Message:

Day 1: Competitive audit complete – [n] platforms analysed

Day 2: User Personas And Information Architecture

Morning (4 Hours): Persona Development

- Continue reading Part A training material: Sections A4, A5, A6
- Create detailed persona cards for all 4 user types (Section A6.1)
- Each persona card must include: name, photo (stock), role, experience level, goals, frustrations, device preferences, accessibility needs, and a quote
- Create empathy maps for each persona

Afternoon (4 Hours): Information Architecture

- Map the complete information architecture of the dashboard using a tree diagram
- Define the navigation hierarchy (primary, secondary, tertiary)
- Create user flow diagrams for the 5 most critical tasks (e.g., alert acknowledgement, VaR drill-down)
- Export all artefacts to PDF and commit

Deliverable:

Persona_Cards.pdf and Information_Architecture.pdf committed to GitHub.

Day 3: Wireframing

Morning (4 Hours): Low-Fidelity Wireframes

- Read Part A: Sections A7 and A8
- Create low-fidelity wireframes for the 4 core screens: Dashboard Home, VaR Analysis, Alert Centre, Settings
- Wireframes should be desktop-only (1440px) at this stage
- Focus on layout, content placement, and information hierarchy – no visual styling

Afternoon (4 Hours): Wireframe Iteration And Review

- Self-review wireframes against the progressive disclosure model (Tier 1/2/3)
- Annotate wireframes with interaction notes (what happens on click, hover, scroll)
- Begin identifying the deliberate errors in Part A training material
- Export wireframes and commit

Deliverable:

Wireframes_Desktop_v1.pdf and Error_Log_Draft.md committed to GitHub.

Days 4-5: Design Token Specification And Component Library Foundation

Day 4 Morning: Design Token Definition

- Set up the Figma file structure: Cover Page, Design Tokens, Atoms, Molecules, Organisms, Templates, Pages, Documentation
- Define all colour tokens (primitive and semantic) as Figma Styles
- Define typography scale and weights as Figma Styles
- Define spacing, border radius, and elevation tokens as Figma Variables

Day 4 Afternoon: Atomic Components

- Build the first 15 atomic components (buttons, inputs, badges, toggles, checkboxes, radio buttons, tooltips, icons)
- Ensure all atoms have all required states (default, hover, active, disabled, focused, error)
- Apply Auto Layout to every component

Day 5 Morning: Molecule Components

- Build 10 molecule components (metric card, search bar, dropdown, tab bar, breadcrumb, alert card, pagination, progress bar, notification toast, empty state)
- Ensure molecules compose atoms correctly via nested instances

Day 5 Afternoon: First Organism And Quality Check

- Build 2 organism components (risk summary panel, data table)
- Run Stark accessibility check on all components
- Document component anatomy on the Documentation page
- Commit Figma file export and component documentation

Deliverable:

Component_Library_v1.fig (or Figma link) with 25+ components committed.

Day 6: Component Library Completion

Morning (4 Hours): Remaining Components

- Build remaining 15+ components to reach the 40-component minimum
- Focus on financial-specific components: Greek display cell, VaR indicator, drawdown meter, spark line, correlation heat map cell, stress test result card

Afternoon (4 Hours): Documentation And Quality Assurance

- Complete anatomy annotations for all 40+ components
- Write do/don't guidelines for 10 key components
- Verify consistent naming convention across all components
- Run full accessibility audit on the component library
- Commit updated library

Deliverable:

Component_Library_v2.fig with 40+ components, full documentation committed.

Days 7-8: High-Fidelity Desktop Prototyping

Day 7 Morning: Core Screens (Desktop)

- Begin high-fidelity design for Screens 1-4 (Login, Dashboard Home, VaR Analysis, Greeks Display)
- Use components from the library exclusively – no detached instances
- Populate with realistic sample data (use the risk metric examples from Part A)

Day 7 Afternoon: Core Screens Continued

- Complete Screens 5-8 (Stress Testing, Drawdown, Alert Centre, Alert Detail)
- Implement interactive prototyping connections between screens

Day 8 Morning: Remaining Screens

- Design Screens 9-12 (Settings, Report Generation, Historical Analytics, Onboarding)
- Ensure role-based view variations are represented (at minimum, show the Dashboard Home in both Junior Analyst and Senior PM configurations)

Day 8 Afternoon: Desktop Prototype Polish

- Implement micro-interactions on 3+ screens (hover effects, loading animations, transition animations)

- Create a clickable prototype flow covering the complete VaR drill-down journey
- Self-review all 12 screens against the design system for consistency
- Commit

Deliverable:

Desktop_Prototype_v1.fig with 12+ interactive screens committed.

Day 9: Desktop Prototype Refinement And Additional Screens**Morning (4 Hours): Refinement**

- Review all 12 desktop screens for visual consistency, spacing accuracy, and data readability
- Implement the alert acknowledgement workflow as a fully interactive flow
- Add realistic data states: loading, empty, error, and populated for at least 3 widgets

Afternoon (4 Hours): Additional Screens (Optional Bonus)

- Design additional screens beyond the minimum 12 if time permits
- Focus on screens that demonstrate depth: What-If Scenario Builder, Monte Carlo Visualization, Regulatory Report Preview
- Commit refined prototype

Day 10: Tablet Responsive Adaptation**Morning (4 Hours): Tablet Layout Strategy**

- Define the tablet layout grid (8-column)
- Determine which desktop components collapse, stack, or transform for tablet
- Adapt Screens 1-6 for 768 x 1024px frame

Afternoon (4 Hours): Tablet Prototype Completion

- Adapt Screens 7-12 for tablet
- Verify touch target sizes (minimum 44px)
- Test interactive flows in tablet prototype
- Run Stark accessibility check on tablet screens
- Commit

Deliverable:

Tablet_Prototype_v1.fig with 12 adapted screens committed.

Day 11: Mobile Responsive Adaptation**Morning (4 Hours): Mobile Layout Strategy**

- Define the mobile layout grid (4-column)
- Apply the adaptive information architecture model from Section A4.3 (mobile shows critical alerts and top-level metrics only)
- Adapt Screens 1-6 for 375 x 812px frame

Afternoon (4 Hours): Mobile Prototype Completion

- Adapt Screens 7-12 for mobile
- Design the mobile-specific lock screen notification (from Case Study 2)
- Implement bottom tab navigation for mobile
- Verify thumb-zone accessibility for primary actions
- Commit

Deliverable:

Mobile_Prototype_v1.fig with 12 adapted screens committed.

Day 12: Usability Testing And Accessibility Audit**Morning (4 Hours): Usability Testing**

- Execute the 5 simulated usability test sessions defined in Section A8.4
- Document findings for each session: task completion rates, time on task, errors, qualitative observations
- Calculate SUS scores for each session
- Create a findings matrix (issue, severity, screen, recommendation)

Afternoon (4 Hours): Accessibility Audit

- Run Stark plugin across all 36 screens (12 per breakpoint)
- Document all violations with screenshots and WCAG criterion references
- Perform manual keyboard navigation test on the desktop prototype
- Write screen reader announcement scripts for 3 key flows (login, VaR drill-down, alert acknowledgement)

- Compile the accessibility audit report
- Commit both reports

Deliverable:

Usability_Test_Report.pdf and Accessibility_Audit_Report.pdf committed.

Day 13: Design Handoff And Specification Document**Morning (4 Hours): Design Specification Document**

- Create the design handoff specification in Notion (or equivalent)
- Document: component API specs, spacing redlines, responsive behaviour rules, animation specs
- Export design tokens as JSON for developer consumption
- Create an asset export manifest

Afternoon (4 Hours): Error Log And Final Documentation

- Finalise the Deliberate Error Identification Log
- For each of the 13 errors: cite the section, quote the error, explain why it is wrong, and provide the correct information with a credible source
- Write the project README.md following Zetheta's metadata standard
- Commit all documentation

Deliverable:

Design_Specification.pdf, Error_Identification_Log.pdf, and README.md committed.

Day 14: Presentation And Client Review Board

Morning (4 Hours): Presentation Preparation

- Create a presentation deck (15-20 slides) covering: project overview, design process, design system walkthrough, key screen walkthroughs, usability findings, accessibility compliance summary, and design handoff approach

Afternoon (4 Hours): Simulated Presentation And Revision

- Self-assess using the Client Review Board rubric
- Upload presentation recording and deck to GitHub

Deliverable:

Presentation_Deck.pdf committed.

Day 15: Sandbox Mode And Final Submission

Morning (4 Hours): Sandbox Mode (Optional Enhancements)

- Implement dark mode variant of the design system
- Add Lottie animations to loading states and transitions
- Design additional screens (What-If Builder, Monte Carlo Viz, Regulatory Report)
- Create a design system documentation site using Zeroheight or Notion

Afternoon (4 Hours): Final Submission

- Perform final quality review across all deliverables
- Ensure GitHub repository is complete with all files
- Verify README.md metadata is accurate
- Transfer repository ownership to ZethetaIntern GitHub account

Final Deliverable:

Complete GitHub repository transferred to ZethetaIntern with all deliverables.

PART E: TOOLS AND PLATFORMS GUIDE

SECTION E1: DESIGN TOOLS

Tool	Purpose	Free Tier Availability	URL
Figma	Primary design and prototyping tool	Free (3 projects, unlimited personal files)	https://www.figma.com/
Adobe XD	Alternative design tool (if Figma unavailable)	Free starter plan	https://helpx.adobe.com/support/xd.html
Stark (Figma Plugin)	Accessibility contrast checker and vision simulator	Free (basic checks)	https://www.getstark.co/
Contrast (Figma Plugin)	Quick colour contrast ratio checker	Free	https://www.figma.com/community
Autoflow (Figma Plugin)	Draw user flow arrows between frames	Free	https://www.figma.com/community
Content Reel (Figma Plugin)	Populate designs with realistic text, images, and avatars	Free	https://www.figma.com/community

SECTION E2: USABILITY TESTING TOOLS

Tool	Purpose	Free Tier Availability	URL
Maze	Unmoderated usability testing on Figma prototypes	Free (1 active project, 10 testers)	https://maze.co/
Useberry	Alternative usability testing tool for Figma	Free (basic plan)	https://www.useberry.com/
Hotjar	Heatmap and session recording (for live prototypes)	Free (35 sessions/day)	https://www.hotjar.com/
Loom	Screen recording for moderated test sessions	Free (25 videos, 5 min each)	https://www.loom.com/
OBS Studio	Open-source screen recording for longer sessions	Free (unlimited)	https://obsproject.com/

SECTION E3: DOCUMENTATION TOOLS

Tool	Purpose	Free Tier Availability	URL
Notion	Design specification documentation	Free (unlimited personal pages)	https://www.notion.so/
Zeroheight	Design system documentation platform	Free (1 styleguide)	https://zeroheight.com/
Google Docs/Slides	Report writing and presentation	Free (Google account)	https://docs.google.com/
Miro	Whiteboarding for information architecture and user flows	Free (3 boards)	https://miro.com/
FigJam	Collaborative whiteboarding within Figma ecosystem	Free (3 boards)	https://www.figma.com/figjam/

SECTION E4: ANIMATION AND MICRO-INTERACTION TOOLS

Tool	Purpose	Free Tier Availability	URL
Lottie by Airbnb	Lightweight animation format for UI micro-interactions	Free (LottieFiles community library)	https://lottiefiles.com/
Rive	Interactive animations and state machines	Free (3 files)	https://rive.app/
Principle	Mac-only prototyping with advanced animations	Free trial (14 days)	https://principleformac.com/
ProtoPie	Advanced interaction prototyping beyond Figma capabilities	Free (basic, 1 pie)	https://www.protopie.io/

SECTION E5: DATA AND FINANCIAL REFERENCE SOURCES

Resource	Purpose	URL
Yahoo Finance	Real market data for realistic dashboard population	https://finance.yahoo.com/
FRED (Federal Reserve Economic Data)	Economic indicators and interest rate data	https://fred.stlouisfed.org/
Investopedia	Financial concept definitions for design context	https://www.investopedia.com/
MSCI Risk Documentation	Understanding institutional risk analytics presentation	https://www.msci.com/
Bloomberg Open Data	Sample financial data sets	https://www.bloomberg.com/professional

SECTION E6: AI-ASSISTED DESIGN TOOLS

Tool	Purpose	URL
Claude.ai (Anthropic)	Design rationale writing, accessibility spec drafting, copy generation	https://claude.ai/
Figma AI (built-in)	Layer renaming, content generation within Figma	https://www.figma.com/
Khroma	AI colour palette generation	https://www.khroma.co/
UIzard	Wireframe-to-design conversion	https://uizard.io/

PART F: SUBMISSION PROTOCOL

Projects are assessed using a comprehensive 1000-point rubric covering seven dimensions: Problem Understanding, Solution Quality, Research & Analysis, Presentation & Clarity, Innovation & Creativity, Feasibility & Practicality and CV alignment. Assessment is conducted through a combination of manual review and AI-powered evaluation.



APPENDIX

This appendix is for assessor reference. It lists all 13 deliberate errors and their corrections. This page should be removed before distribution to interns.

Error #	Section	Error Description	Correct Information
1	A1.1	States Nielsen's 8 Usability Heuristics	Nielsen's 10 Usability Heuristics (not 8)
2	A1.2	States Fitts's Law was formulated in 1964	Fitts's Law was published in 1954 (Fitts, 1954)
3	A1.3	States VaR satisfies all four coherence properties	VaR does NOT satisfy sub-additivity; only CVaR is coherent
4	A1.3	Sharpe annualisation: $\sqrt{T}$ applied to ratio directly	$\sqrt{T}$ is applied to Sharpe only when returns are computed at sub-annual frequency; the standard formula annualises return and vol separately
5	A2.2	States categorical palette for correlation heat maps	Diverging palette should be used (not categorical) since correlation has a meaningful zero centre
6	A2.5	States 14px minimum applies to ALL text including footnotes	WCAG 2.1 AA minimum is 12px for body text; 14px minimum is a recommendation, not a strict requirement. Footnotes and axis labels commonly use smaller sizes
7	A3.4	States 6-pixel base grid is industry standard	Both Material Design 3 and Apple HIG use an 8-pixel (not 6-pixel) base grid system
8	A4.3	Recommends CSS transform: scale() for mobile adaptation	Scaling down the desktop view is a universally condemned anti-pattern. W3C Mobile Web Best Practices explicitly recommend against this approach

9	A5.3	States 4.5:1 contrast ratio required between adjacent data series	WCAG 1.4.11 requires 3:1 contrast against adjacent colours for non-text elements, not 4.5:1. The 4.5:1 ratio applies to text only
10	A6.2	States client-side cookie-based role management is secure	Client-side cookies are trivially editable by users. Role-based access must be enforced server-side. The Figma prototype approach of separate screens per role is also inefficient; use component variants with role-based visibility props
11	A7.4	Recommends all alerts at P1 Critical by default	This is the exact opposite of best practice. Alert fatigue is combated by proper severity classification, not by making everything critical. Audio on all alerts also worsens fatigue.
12	A8.2	Recommends leading, directive usability test tasks	Leading tasks invalidate test results. Tasks must be open-ended and scenario-based, not directive. This is a fundamental principle of usability testing methodology.
13	A8.4	States SUS has 15 questions on 7-point scale, developed by Jeff Rubin in 1996	SUS has 10 questions on a 5-point Likert scale (scoring 0-100), developed by John Brooke in 1986 at Digital Equipment Corporation

SAMPLE DATA FOR DASHBOARD POPULATION

Use the following sample data to populate your dashboard prototypes with realistic figures:

Portfolio Summary Data

Metric	Value	Change (1D)	Status
Total Portfolio AUM	USD 2.4 Billion	+0.3%	Normal
1-Day 95% VaR	USD 12.8 Million	+1.2M	Warning (90% of limit)
1-Day 99% CVaR	USD 28.4 Million	-0.6M	Normal
Maximum Drawdown (YTD)	-8.7%	-0.4%	Normal
Active Alerts	7 (2 Critical, 3 High, 2 Medium)	+3	Critical
Sharpe Ratio (90-day)	1.42	-0.08	Normal
Daily P&L	+USD 3.2 Million	N/A	Positive
MTD P&L	+USD 18.7 Million	N/A	Positive
YTD P&L	+USD 124.3 Million	N/A	Positive

ZeTheta
ADD EXPERIENCE TO KNOWLEDGE

Greeks Summary (Equity Options Book)

Greek	Portfolio Level	Limit	Utilisation
Delta	+450,230	+/- 500,000	90.0% (Warning)
Gamma	-12,450	+/- 20,000	62.3% (Normal)
Theta	-USD 180,400/day	N/A (Monitored)	N/A
Vega	+USD 2.1M per 1% vol	+/- USD 3M	70.0% (Normal)
Rho	-USD 340K per 1% rate	+/- USD 500K	68.0% (Normal)

Stress Test Results

Scenario	Portfolio Impact	Worst-Hit Asset Class	Recovery Estimate
2008 GFC Replay	-USD 312M (-13.0%)	Credit (High Yield)	6-12 months
COVID March 2020	-USD 198M (-8.3%)	Equities (EM)	3-6 months
Interest Rate +200bps	-USD 89M (-3.7%)	Fixed Income (Long Duration)	2-4 months
Equity Crash -20%	-USD 224M (-9.3%)	Equities (US Large Cap)	4-8 months
EM Currency -30%	-USD 156M (-6.5%)	EM Local Currency Bonds	6-12 months

FIGMA FILE STRUCTURE TEMPLATE

Set up your Figma file with the following page structure:

1. Cover: Project title, intern name, date, version number
2. Design Tokens: Colour palette, typography scale, spacing system, elevation, icons
3. Atoms: All atomic components with variants and states
4. Molecules: All molecule components
5. Organisms: All organism components
6. Templates: Page-level layout templates for each breakpoint
7. Pages - Desktop: 12+ screens at 1440 x 900px
8. Pages - Tablet: 12+ screens at 768 x 1024px
9. Pages - Mobile: 12+ screens at 375 x 812px
10. Prototype Flows: Interactive connections and flow annotations
11. Documentation: Component anatomy, do/don't guidelines, usage notes
12. Exploration: Rejected concepts, iteration history, design rationale notes



PRESENTATION STRUCTURE GUIDE

The Client Review Board presentation (Day 14) must follow this recommended structure:

- 1. Introduction (2 minutes):** Introduce yourself, the project brief, and the key design challenge you addressed.
- 2. Research and Discovery (3 minutes):** Summarise your competitive audit findings, persona development, and information architecture decisions. Show 2-3 key insights that shaped your design direction.
- 3. Design System Walkthrough (3 minutes):** Present your design token specification, demonstrate the component library with 5-6 key components, and explain your naming convention and organisational logic.
- 4. Desktop Prototype Demo (4 minutes):** Walk through the 3 most important user flows on the desktop prototype: (a) Portfolio Overview and VaR drill-down, (b) Alert acknowledgement workflow, and (c) Role-based view switching between Junior Analyst and Senior PM.
- 5. Responsive Demonstration (2 minutes):** Show 2-3 screens on tablet and mobile, explaining the adaptive information architecture decisions you made for each breakpoint.
- 6. Usability and Accessibility (3 minutes):** Present your top 3 usability findings, demonstrate how you addressed them, and show your WCAG compliance summary. Highlight any particularly challenging accessibility problems you solved.
- 7. Design Handoff (1 minute):** Briefly show the specification document structure and explain how a developer would consume your deliverables.
- 8. Q&A (2 minutes):** Prepare for questions from each stakeholder persona: Maya (CEO) will ask about competitive differentiation, Raj (CTO) will ask about performance and implementation feasibility, Elena (Compliance) will ask about WCAG specifics, and David (Engineer) will ask about component API clarity.

END OF PROJECT DOCUMENT